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The Ethics of Information Systems: The Present State of the Discussion and Avenues for Future Work

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THE ETHICS OF INFORMATION SYSTEMS: THE PRESENT STATE OF THE DISCUSSION AND AVENUES FOR FUTURE WORK

Research in Progress

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Abstract

Despite occasional pleas to incorporate the ethics of information systems into the field's research agenda, the analysis of the moral ramifications of information systems has remained unsystematic in nature. Work on a remarkable variety of ethical issues has appeared, but the field has yet to produce a coherent conception of how it construes and studies the ethics of information systems. In this research-in-progress paper, we report on work to redress the situation by addressing three goals. First, we provide an overview of contributions made to the study of ethics in information systems research in recent decades. Second, we present a classificatory scheme to organize past and future work on the subject. Third, we identify themes that have received little attention to date, proposing routes for future work. In particular, we find that our field has yet to fully explore the moral implications of the very artifacts of its own design. We conclude with a call to action for information systems researchers to work towards a structured account of ethics as part of the field's research agenda.

Keywords: Ethics, Moral Philosophy, Information Systems, Methodology, Research Agenda.

1 Introduction

There is no foreseeable end to the expansion of influence exerted by information systems (IS) on almost all areas of personal, organizational, and societal life. IS are deployed in present-day organizations to support or even execute activities of all orders, from operational tasks and administrative practices to strategic planning. Consequentially, information systems have become significant catalysts of *moral* or *immoral* behavior of individuals, groups, and organizations.

The field of information systems research (ISR) takes as its task the study of IS in profit and non-profit organizations. In pursuing this task, however, the study of the *moral implications* of IS has for a long time received only scattered attention. In the early days of ISR, ethical concerns were of earnest interest only to few investigators, such as Mumford (1983) in her classical approach to humanistic systems design, and Mason (1986) in his early information ethics framework. More than two decades ago, then, Laudon diagnosed that “Despite the explosion in information technology [...], scholars, students,

and practitioners would be hard-pressed to claim similar progress in ethical thinking about information technology” (Laudon, 1995, p. 33). In the years that followed, a modicum of work on ethics in ISR appeared. Yet, a decade later, Walsham still felt compelled to urge in a research commentary that “if the IS field is to inspire in the future, I wish to argue that the field needs a strong ethical agenda” (Walsham, 2012, p. 89). Similarly, Adam concluded that “Looking back over the last 15 or so years it is clear that, whatever turns IS did make, the ethical turn was not one of them” (Adam, 2012, p. 102).

Since then, IS researchers have begun to take serious the ethical dimension of IS, as is evidenced by the growing number of special issues and conference tracks on the theme. Attention has been directed at a considerable number of topics, covering such diverse areas as knowledge management (e.g., Chatterjee and Sarker, 2013), healthcare systems (e.g., Dadgar and Joshi, 2018), the dark side of the internet (e.g., Benjamin et al., 2019), and consumer behavior (e.g., Tsai et al., 2011). However, despite, or because of, the concern with the many individual aspects of ethics and IS in specialized fields, efforts have remained somewhat unsystematic in nature. There is as yet no coherent and widely accepted conception of how the field understands its ethical agenda, and what it takes to be covered by that agenda. One fundamental problem with the lack of such an overarching conception is that conceals blind spots: it is not apparent where the field overlooks problems that fall into its very own area of responsibility.

This research-in-progress paper reports on work to redress the situation by addressing three goals. First, we intend to provide an overview of the range of contributions that have been made to the study and methodological integration of ethics in ISR in the past decades. Second, we seek to present a broad framework to classify existing work and identify neglected areas. Both goals will be addressed by means of a *mapping study*. The analysis is intended to contribute to a future conception of a domain ethics of ISR. Third, on this basis, we wish to suggest avenues for future research.

The paper proceeds as follows. In the next section, we provide some theoretical background on the realm of ethics. In the third section, we cover related work. Section 4 explains the strategy applied in conducting the study, and Section 5 presents the main contribution of the paper, a map of extant work on ethics in ISR. Following a discussion in Section 6, the paper closes with Section 7.

2 Ethics: Background and Domain-Specific Subfields

Ethics, or moral philosophy, is one of the fundamental branches of philosophy. It is also one of the oldest branches, with roots extending far back at least to Greek antiquity (Schwemmer, 2005). According to one dominant conception, ethics is essentially the study of *morals* or morality (Deigh, 1999; Schwemmer, 2005). Morals and morality are usually thought of as systems of *norms* or *principles* directing human action in the light of specific qualities (ibid). Among the major moral qualities most commonly studied are *goodness*, *rightness*, and *justness* (ibid; see also Crisp, 2000).

Ethics has been divided along several lines. One important distinction is into *metaethics*, *descriptive ethics*, and *normative ethics*. Metaethics is concerned with the ontological and epistemological status of the objects of ethics *themselves* (Deigh, 1999). Descriptive ethics analyzes *actual* moral systems of different groups and cultures. Normative ethics, in turn, “is concerned with the *articulation* and the justification of the fundamental principles that govern the issues of [...] what we morally ought to do,” that is, moral systems (Driver, 2005, p. 31). Another important, albeit controversial, distinction is into *theoretical* and *applied* ethics (Crisp, 2000; Gethmann, 2005). Applied ethics, unlike theoretical ethics, is closely concerned with questions of morality in concrete situations and contexts of public life. However, applied ethics still requires abstract concepts and justifications; the two, therefore, cannot be separated entirely (Gethmann, 2005). What is important for our purposes is that the field of applied ethics is often itself divided into several areas. Classical examples are *medical ethics* and *business ethics*. These subfields are often called *domain ethics*, and it is here that an ethics of IS could place itself.

Several domain ethics have been established in recent decades. Some of them, in fact, are already intimately related to ISR. Among the domain ethics most closely related are technology ethics, information ethics, and business ethics (Bendel, 2015, pp. 741ff.). *Technology ethics* is concerned with the moral use and production of technology of any kind (Bendel, 2015). Although the field has yet to pro-

duce a well-defined agenda, it basically analyzes technology in terms of its impacts on political, cultural, and working life (Franssen et al., 2018). Analogously, *information ethics* studies the ethical questions involved in the use of computerized information technology (IT) (Bendel, 2015). It examines the effects of IT on human life, relationships, and politics (Bynum, 2018). *Business ethics*, in turn, is concerned with the moral issues arising from economic transactions among individuals and organizations, as well as questions of how to govern such transactions in the first place (Moriarty, 2017).

These existing domain ethics already cover important objects of study which also fall into the scope of ISR, namely, information technology and economic conduct. However, none of these domain ethics are specifically devoted to the core subject of our discipline, that is, the *development, use, and maintenance of information systems in and for the social action systems of organizations*. This subject cannot be covered by a perspective focusing either on the information technology or the business side alone. Conversely, however, one must be careful not to assume that the mere presence of an IS makes a moral problem a problem of ISR. Often, IS figure only secondarily in moral problems relating to other fields, as when an IS appears incidentally in a case of a medical life-or-death decision. Working out these and other details and boundaries of a *domain ethics of IS* is what we consider a vital responsibility of ISR. What follows is an overview of some of the previous steps that have been made towards such a conception, and some of the future steps it would seem profitable to make.

3 Related Work

Several other authors have already provided overviews of work on ethics in ISR. One noteworthy example is the study by Stahl (2012). He classified research on ethics in ISR until 2012 with respect to the type of research and so-called levels of normativity (*viz.*, reflection and metaethics, ethical theory, explicit morality, and moral intuition). In contrast to Stahl, our overview classifies existing work in relation to themes and aspects of ISR, such as type of ISR artifact and outcome. This has the advantage of revealing actual areas of IS research in need of more careful appraisal.

Another overview has been offered by Paradice et al. (2018). Building on a literature review, they have classified work on ethics in ISR into varied subjects at different levels of abstraction. In contrast to Paradice et al. (2018), we focus more on the structure and scope of studied topics instead of summarizing the discussion. Besides these explicit review studies, there are, as might be expected, various authors who begin their articles on individual aspects of ethics in ISR with brief surveys of previous work. This is the case, for example, with the papers by Davison (2000), Smith (2002), and Mingers and Walsham (2010). However, unlike our paper, none of these publications are expressly intended to offer an overview of the available literature and points of emphasis of the debate.

4 Approach to the Study

In this section, we explain the approach taken in our study of work on ethics in ISR, covering the *aim and scope* (Section 4.1), the *type of study* (Section 4.2), and the *specific strategy* applied (Section 4.3).

4.1 Aim and Scope

The central aims of the following study are the two goals stated in the introduction, namely, to provide an *overview of contributions* to the study and the methodological integration of ethics in ISR, and to obtain a broad *classificatory scheme* to organize existing and possible future work. Both of these aims are intended to contribute to the ultimate goal of establishing *conception of a domain ethics of IS*; the first aim, by yielding an impression of the present state of the discussion, and the second aim, by giving direction and coherence to coming efforts. As regards the contributions considered, focus is placed specifically on written materials that at least partly address themselves, from an *abstract* or even expressly philosophical or methodological perspective, to the place, role, and scope of ethics in ISR. For example, research commentaries are paradigm cases of such work. In contrast, the aim is not to cover all papers dealing with specialized facets of ethics in ISR. The fundamental reason for this restriction

is, again, the ultimate goal pursued. Developing a conception of a domain ethics of IS requires to deal, in abstract and general terms, with the objects, boundaries, and relations of and between ethics and ISR, and therefore we concentrate on work assuming precisely such abstract and elevated a stance.

The scope of our study is work in ISR in the last three or so decades. We do not consider work in adjacent disciplines, like economics and computer science, as we deliberately wish to stimulate discussion on the ethics of our field. Furthermore, the present study is not and cannot be exhaustive in any way. For one thing, it is beyond the scope of a single article to screen all contributions of a complete discipline. What is more, the theme of ethics is particularly delicate in that it is often treated alongside other topics, without being raised to the status of a primary subject. Discussions of ethics often occur without ever mentioning terms such as ‘ethics’ or ‘morality’, relying instead on value-laden terms like ‘justice,’ ‘safety,’ or ‘harm.’ Accordingly, what we intend to provide is a broad overview, or tentative map, not an exhaustive catalog. For this reason, we have followed the strategy of a *mapping study*.

4.2 What is a Mapping Study

A mapping study is a type of review study, which is conducted to identify and categorize the existing literature in a particular field of study (Petersen et al., 2008; Petersen et al., 2015). It provides a broad overview of a research area and can be used to identify gaps in the existing literature. Mapping studies are also useful in identifying topic areas within a field where enough primary studies exist to justify a more in-depth review of the theoretical developments (Kitchenham et al, 2011).

Mapping studies are broadly related to structured literature reviews (SLR), but they differ in several ways. In mapping studies, the goal is to review a broad topic area and to classify the primary research papers, whereas SLRs aim to answer narrower questions pertaining to a limited literature pool (Kitchenham et al., 2012). The guiding questions of SLRs, in turn, determine the process of identifying the primary research papers, direct the extraction of data, and facilitate the synthesizing of the extracted data. Another important difference is that, on one hand, SLRs seek to synthesize primary studies taking into account the research outcomes and, on the other, mapping studies try to classify the existing literature according to newly conceived abstractions (Kitchenham et al, 2011).

4.3 Strategy and Key Decisions

The procedures on which mapping studies are based can be varyingly mechanistic. The choice of procedure usually depends on the objectives of the study (Cooper, 2016). While many mapping studies follow the review protocols of SLRs, we adopted a flexible literature selection process and a discursive collaborative review procedure. In brief, this is because (1) there is considerable *variation* in terminology surrounding the field of ethics; (2) discussions of ethics often occur *incidentally* alongside discussions of other topics; and (3) many discussions of ethics appear in publications whose title *does not contain* surface terms like ‘ethics’ or ‘morality.’ As a consequence, any search procedure based on rigid search strings alone would inevitably fail to retrieve relevant work—for example, the important commentaries on ethics as part of the IS agenda by Adam (2012) and Walsham (2012).

Our search strategy, therefore, was threefold. First, we searched for relevant work using such obvious keywords as ‘ethics’ and ‘moral’ in combination with ‘information systems.’ The database used for all keyword-driven activities was Google Scholar. Second, we analyzed the bibliographies of recorded items for further relevant publications, repeating the process until no more new titles were found. We have relied on backward search exclusively, because the research commentaries of interest to the present study tend to be cited in a broad variety of contexts unrelated to our subject. Third, we added one relevant title of which we were aware because of our general familiarity with the German IS community (namely, Fehling and Jahnke, 1999). In line with our stated aims (see Section 4.1), the selection criteria applied were (1) whether a title takes ethical implications of IS as its *principal object*, and (2) whether a title explicitly discusses, from a *high-level viewpoint*, the place of ethics in ISR.

To develop the classificatory scheme, we employed a discursive, iterative procedure. Essentially, we studied the various publications, we identified and discussed their primary objects as well as the type

of research, and we successively derived and refined a classificatory system with two levels. In the process, we applied acknowledged practices. Each piece of literature was independently reviewed by at least two authors. Also, we conducted several reviewer agreement rounds, where we jointly reflected on inter-reviewer reliability issues (Belur et al., 2018) and resolved cases where two or more reviewers disagreed over the categorization of a paper. The categories themselves, ultimately, have two sources. The first source were the papers under consideration, from which categories have been derived by way of abstraction. The second source was the authors' general knowledge of ISR and its subfields. Essentially, it is only the second category listed below which is rooted in the authors' knowledge, whereas all others have been derived from the papers themselves. Still, the resulting scheme is our own conceptual abstraction, and divergent conceptualizations can be defended as well.

5 A Map of Work on Ethics in Information Systems Research

In this section, we present the results of our literature study, providing, in effect, an initial cartography of work in ethics in ISR. Specifically, for one, we present the developed *classificatory scheme* for work on ethics in ISR. Simultaneously, we present how *extant publications* fall into the categories of this scheme. Our scheme involves three main categories: (1) *Domain ethics in IS*, (2) *Ethics of IS design artifacts*, and (3) *Ethics in doing ISR*. All three categories are further refined into several subcategories. The categories are non-exclusive, so that recorded publications can fall into several categories at once. The overall results are summarized in Table 1. A checkmark indicates a fit between a reference and a category; a checkmark in brackets indicates a partial fit.

5.1 Domain Ethics in IS

The first main category contains work on any type of ethically laden behavior or issue related to the *use, maintenance, and development of information systems*. It consists of four subcategories.

The subcategory "ethical behavior associated with IS in general" describes research dealing with the morals of information technologies in general, treating them in a *uniform* and undifferentiated manner (i.e., "computer technologies" as a whole). Some publications investigated (un)ethical types of behaviors in an indiscriminate way, without distinguishing different forms of behavior (e.g., Cohen and Cornwell, 1989; Walsham, 1996; Banerjee et al., 1998; Li et al., 2014; Benjamin et al., 2019), while others investigated specific types of behavior such as file-sharing (e.g., Hansen and Walden, 2013).

The subcategory "moral theory and ISR ethics" is of great significance in view of the motivation of this paper. In this subcategory, IS researchers have already begun to integrate *moral theories* with the IS agenda, or otherwise to develop *theoretical conceptions* of an ethics of ISR. Chatterjee and colleagues have done much work in this area, devising, for example, a theoretical synthesis of the objects of moral philosophy and knowledge management (Chatterjee and Sarker, 2013), or applying postmodernist thought to IS systems development (Chatterjee et al., 2009). Stahl (2012), as another example, proposes an abstract, multi-level framework linking ISR to moral philosophy.

The next two subcategories are related in that the first contains the second. The subcategory "ethical behavior associated with specific types of IS" covers research dealing with morally laden applications of special forms of IS and related technologies. As already emphasized at the outset, in recent decades IS researchers have done much work of this character. Objects of study include health management IS (Dadgar and Joshi, 2018), knowledge management IS (Chatterjee and Sarker, 2013), and search engines (Tsai et al., 2011). Research methods range from argumentative studies, to qualitative studies, to experimental studies. A particularly extensive strand of research of this kind is concerned with "privacy, data security, and property rights". To highlight its special status, we have included it as a distinct subcategory. Contributions in this subcategory, for example, investigate privacy (Gal-Or et al., 2018), file-sharing (Hansen and Walden, 2013), and the darknet (Benjamin et al., 2019). It is important to stress, however, that the recorded references of the last two subcategories make up merely a fraction of related work; they stand, *pars pro toto*, for specialized research on IS ethics at large.

Reference	Type	Domain Ethics in Inform. Systems				Ethics of IS Design Artifacts			Ethics in Doing IS Research				
		Eth. Behavior (General)	Moral Theory and ISR Ethics	Eth. Behavior (Specific)	Privacy, Data Security, Prop.	Software Developm. Artifacts	Soc.-Techn. Design	Design Options	Research Methods	(Meta-)Observations	Ethics of IS Researchers	Teaching	Calls to Action
Cohen and Cornwell, 1989	e	(✓)										✓	
Couger, 1989	t											✓	
Laudon, 1995	t	✓											✓
Lewellyn, 1996	n										✓		
Smith et al., 1996	n	✓			✓								
Walsham, 1996	c	✓	✓							✓			
Banerjee et al., 1998	n	✓											
Fehling and Jahnke, 1999	c						(✓)			✓			✓
Smith and Hasnas, 1999	c	✓		(✓)									
Davison, 2000	c								✓	✓			
Davison et al., 2001	c								✓	✓			
Smith, 2002	t		✓										
Chatterjee et al., 2009	t		✓										
Mingers and Walsham, 2010	l		✓	✓					✓	✓			
Tsai et al., 2011	e			✓	✓			✓					
Adam, 2012	c	(✓)								✓			(✓)
Berente et al., 2012	t		✓						✓				
Pouloudi, 2012	c												✓
Stahl, 2012	l		✓							✓			
Walsham, 2012	c	(✓)							✓	✓			(✓)
Chatterjee and Sarker, 2013	c		✓	✓									✓
Hansen and Walden, 2013	e	✓		✓									
McBride, 2014	c				(✓)				✓	✓			
Li et al., 2014	n	✓											
Bendel, 2015	c	✓											✓
Chatterjee et al., 2015	n	✓	✓										
Lowry et al., 2015	c	✓			✓				(✓)				✓
Dadgar and Joshi, 2018	q			✓		(✓)							
Gal-Or et al., 2018	t				✓								
Paradice et al., 2018	l									✓			
Benjamin et al., 2019	t	✓		✓					✓				✓

Types: e: Experimental quantitative empirical; n: Non-experimental quantitative empirical; q: Qualitative empirical; t: Theoretical-conceptual; l: Literature review; c: Commentary.

Classification: ✓: Fit; (✓): Partial fit.

Table 1. Classificatory scheme and classification of selected titles.

5.2 Ethics of IS Design Artifacts

The second main category “ethics of IS design artifacts” covers work on *artifacts* guiding or otherwise bearing on the design of IS and their ethical implications. This includes methods, modelling methods, process models, design frameworks, and similar items. Because artefacts of these kinds are among the classical results of design-oriented research in ISR, this category concerns the analysis of the ramifications of the *products of IS research* itself. Remarkably, we identified only few contributions expressly attending to this purpose. The category consists of three subcategories.

First, the subcategory “software development-related artifacts” concerns research which deals, under consideration of ethical norms, with the design and evaluation of artifacts for the *development of the technological part of IS* (such as software development methods). Only one title falls into this subcategory. Dadgar and Joshi (2018) have investigated the ethical impact of a diabetes management system using value sensitive design. Second, “socio-technical design” refers to the study of the moral ramifications of artifacts for the analysis and design of *socio-technical systems* (such as enterprise modeling methods). Stunningly, only one paper peripherally touches upon this area, to wit, the old paper by Fehling and Jahnke (1999). Finally, “empirical study of design options” refers to work that *evaluates high- or low-level prototypes* from which knowledge about the ethically proper and improper design of IS artifacts can be derived. Again, only one title fits this subcategory. Tsai et al. (2011) investigated how users’ purchasing behavior is affected by details of the privacy standards of a website.

5.3 Ethics in Doing IS Research

The last major category “ethics in doing IS research” describes publications that deal with ethical issues bound up with the process of *information systems research itself*. This can range from ethical problems related to research methods, to ethical aspects of teaching ISR, to calls to action.

The subcategory “research methods” covers papers on the *methodology of ethics in ISR*. This subject has been investigated by numerous authors. Contributions in this subcategory either take a practical perspective, discussing, for example, codes of ethics (e.g., Davison, 2000), or they discuss theoretical approaches to the analysis of moral matters (e.g., Mingers and Walsham, 2010). Then, there is a varied set of papers fitting the subcategory “(meta-)observations on the field and community of ISR”, which analyze and comment on what research is *actually done* in the IS community. Whereas some contributions have canvassed the ethical topics actually treated in ISR (Paradice et al., 2018), others have focused on the type of ethical theories used (Stahl, 2012; Mingers and Walsham, 2010; McBride, 2014). Still other papers looked at the codes of ethics applied (Walsham, 1996), professional ethics (Davison, 2000; Davison et al. 2009), and codes of practice (Davison et al., 2001).

Work in the subcategory “actual ethics of IS researchers” describes empirical studies on moral convictions where *IS researchers or professionals are the subjects*. Closely related, the category “teaching IS ethics” deals with work on ethics as part of the *IS curriculum*. Only few references fall into these subcategories. The moral views entertained by IS academics were investigated in an early study about different sorts of unethical behaviors (Lewellyn, 1996). Another early study in the teaching domain looked at whether the teaching of ethics can improve students’ moral attitudes (Cohen and Cornwell, 1989). Equally early on, Couger (1989) gave a personal account of teaching morality in ISR. Finally, a number of “calls for action” have been published, forming a subcategory that includes every proposal or commentary directed at IS researchers, encouraging them to devote attention to ethical questions. It is here that some of the most pronounced calls for an incorporation of an ethics of IS into the ISR research agenda are found (Laudon, 1995; Walsham, 2012; Adam, 2012). Besides these important calls, titles in this subcategory are quite diverse, ranging from very specific commentaries in the area of cyborgs (Pouloudi, 2012) to more general suggestions in the privacy area (Lowry et al., 2017).

6 Discussion

Relatively few abstract and general accounts of the place, role, and scope of ethics in ISR. The foregoing literature overview has demonstrated that an emphatic and welcome stream of work on ethics in ISR has appeared in recent decades, particularly when it comes to the analysis of specialized ethical issues, like privacy, data security, and property rights. Despite this, it is fair to say that the type of work which has been the major interest of this study, viz., abstract or even designated methodological contributions on the role of ethics in ISR, has remained fairly sporadic and isolated to this day. It is the case that several authors have made valuable contributions toward a general conception of ethics in ISR (e.g., Smith, 2002; Chatterjee et al., 2009; Mingers and Walsham, 2010; Berente 2011). But in comparison to the wealth of articles addressing themselves to specific aspects, studies taking a broader

perspective have remained limited, with regrettable results (see below). We would, therefore, like to prompt IS researchers to expand and solidify work on the general features of a domain ethics of IS.

No unified conception of ethics of IS. The most important finding, and a direct consequence of the previous point, is that today there is nothing even remotely resembling a widely shared conception of the ethics of IS. Studies of individual ethical issues usually fashion specialized theoretical lenses, without concern for the principal scope and structure of moral theorizing in ISR. The relatively few publications which do work toward such a common conception (cf. above), in their turn, only take initial steps and, still more unfortunately, are rarely taken up by other authors. There may be many reasons for this lamentable state of affairs, but we feel that the field's tendency to require a (supposedly) 'original' theoretical contribution of every paper (Avison and Malaurent, 2014) has a share in it. One danger associated with the absence of a unified view is that studies of ethics in ISR lack consistency in direction; another danger is that they remain incoherent or altogether disconnected. Still another risk, elaborated below, is that ISR overlooks ethical problems falling into its own scope of responsibility. Developing a conception of the ethics of IS entails several tasks. It is necessary, for example, to analyze and structure the many ways in which IS may have impacts of ethical significance; it is necessary to study these aspects of IS in relation to classical and modern moral theories; and it is necessary to integrate the resulting framework and the attendant modes of analysis into the methodology of our field.

Little work on design artifacts of ISR. A more specific finding is that little work has been done on the ethical evaluation of the artifacts designed by IS researchers. This is surprising and worrisome. The design of artifacts, such as methods, modeling languages, prototypes, and reference architectures, is one of the traditional hallmarks of ISR, and it is through them that ISR, ultimately, molds possible future worlds (Frank, 2017). For example, the concepts of modeling methods for decision making and problem solving inevitably shape what is, and what is not, recognized as part of a problem (Bock, 2016). In the case of, say, the problem of whether or not to outsource a firm's IT services (see *ibid*), ISR design artifacts, thus, directly affect other people's lives. We, therefore, suggest that another avenue for future work should consist in a study of how design-oriented research methods in ISR can be augmented with components to critically assess the ethical implications of the newly devised artifacts.

Little empirical work. Another finding is that empirical, and particularly experimental, work on ethics in ISR has remained limited. In large measure, of course, this is natural. Only the fundamental questions of descriptive ethics, but not those of normative ethics and metaethics (see Section 2) are amenable to empirical methods. Nevertheless, empirical studies can contribute to an understanding of whether systems are actually used as intended, and whether they do promote or discourage moral or immoral conduct. We, therefore, consider further work of this nature another valuable route.

7 Conclusion

In this paper, we have sought to make a contribution to establishing an ethics of ISR. We have, firstly, presented a classification scheme for existing work on ethics in ISR; we have, secondly, classified some of the salient publications in recent decades using this scheme; and we have, thirdly, identified some routes for future research based on the findings of our literature study. Of course, given the restricted scope of the paper, our work must remain limited in several ways. For one thing, as stressed throughout, our literature survey is by no means exhaustive. Moreover, we could not even begin to discuss in substantive terms what steps have previously been made towards the conception of an ethics of IS and how one might expand upon that. These tasks are part of our ongoing research.

We hope that the classification scheme can provide an orientation for IS researchers who are as yet unfamiliar with moral theory in ISR, just as we hope that it will be instrumental in directing attention at as yet underemphasized areas, like the study of the moral consequences of the very products of IS research. Most importantly, we hope that the results will stimulate future work on a coherent and unified understanding of what an ethics of ISR can be, what it involves, and how it links up with classical moral theories in philosophy. We are convinced that this will help increase the impact of valuable previous research on ethics in ISR, just as it will help discover novel areas worthy of illumination.

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